

Elijah Akpan

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Education

Worcester Polytechnic Institute | Worcester, MA

Exp. Graduation Date: 2028

B.S/M.S in Mechanical Engineering

GPA: 3.6

Relevant Coursework: Thermodynamics, Differential Equations, Advanced Engineering Design, Dynamic Systems, Linear Algebra

Skills

Softwares: SolidWorks, MasterCAM, AutoCAD, LabView, CATIA, KiCAD, Vericut, Vigrade, Siemens NX

Programming Languages: MATLAB, Python, SCPI, G-code

Hardware: Arduino, Haas CNC Machines, BambuLab, Raspberry Pi, Prusa, Ultimaker

Professional Experience

Mechanical Engineering Intern, **Polamer Precision** | New Britain, CT

May 2026 - Aug 2026

- Developed and optimized manufacturing processes for engine components used in 20000+ horsepower aerospace engines.
 - Programmed complex CAM toolpaths using MasterCAM for engine components in both turning and milling.
 - Reduced production errors by optimizing machine and CAM simulation through creating machine templates for Vericut.
- Reverse engineered VTL machines using SolidWorks and wrote g-code/python macros for tool changing and turret actuation.

Mechanical Engineering Intern, **TTM Technologies** | Syracuse, NY

Aug 2025 - Dec 2025

- Designed and tested RF components and PCBs for fighter jets.
- Help create design of a new 3dB hybrid coupler using thermal FEA simulations to maximize both thermal and RF performance.
- Optimized quality process for RF parts by saving engineers 100+ hours of quality work by developing a MATLAB GUI that uses shape recognition to automatically detect defected vias from folders containing 200+ image.

Autonomous Vehicle Researcher, **Autonomous Vehicle Mobility Institute** | Worcester, MA

Oct 2024 - Present

- Design and manufacture components for an autonomous vehicle laboratory that specializes in off-road vehicles.
- Led the design and manufacturing of a specialized vehicle simulation cave by modifying T-slots, sourcing plywood for screen, machining projector mount, and designing PLA brackets used for support. The lab secured a \$65,000 dollar grant due to the cave.
- Helped upgrade a \$100,000-dollar unmanned ground vehicle by designing and manufacturing the suspension system, mounting bracket for DC motors, and the LiFePO4 battery cage.

Teaching Assistant, **Worcester Polytechnic Institute, Washburn Labs** | Worcester, MA

Feb 2025 - Present

- Lead 2 lab sections consisting of 30+ students on how to manufacture their own 10 piece Stirling Engine using MasterCAM & Haas Mini Mills/Lathes.
- Show students how to calculate RPM, Feed rate, and other critical manufacturing values to program and machine each part of the Sterling Engine.

Extracurriculars

Performance Lead, **WPI Formula SAE** | Worcester, MA

Aug 2024 - Present

- Leading the development of a 230 horsepower electric FSAE racecar through optimizing the powertrain and the vehicle dynamics using CAE, CAD, and CNC machining.
- Led the development of a polycarbonate enclosure with a \$1000 dollar budget that protects, cools, and simultaneously charges a total of 8 highly explosive lithium ion segments that generate a combined 400 volts.
- Designed and manufactured a pedal box that increased throttle map by 10%. Responsible for 3D design, CAM, and machining of the entirety of the pedal box.

Projects

Engine Simulator For Toy Ride On Cars

July 2025 – Present

Invented a (PATENT PENDING) engine simulator for ride on cars that replicates engine noises of the real car based on pedal tracking. Raised \$2000+ dollars so far for the invention.